MIL-C-24643/40B
3 October 1986
SUPERSEDING
MIL-C-0024643/40A(SH)
8 November 1985
MIL-C-24643/40
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SMU (INCLUDING VARIATION LS1SMA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS1SMU (unarmored) and LS1SMA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, in accordance with ASTM B 286, size 22-7, tin coated. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness 0.030 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth - Braided shield of AWG no. 36, tin coated copper. Braid angle of 20 to 25 degrees; minimum coverage of 90 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.
Sixth - The required number of singles (see table I) cabled around a center filler with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1. Fillers shall be employed to form a firm, round cable.
Seventh - Binder tape applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.

Construction variant for type LS1SMU

Eighth - Cross-linked polyolefin jacket. Minimum thickness 0.050 inch. Cable surface marking required.

Construction variant for type LS1SMA

Eighth - Cross-linked polyolefin jacket. Minimum thickness 0.050 inch.
Ninth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/40	Size 2/	Number of conductors	Diameter over completed conductors nominal (inch)	Overall diameter (inches)			
				Type LS1SMU		Type LS1SMA	
				minimum	maximum	minimum	maximum
-01UN	5	5	0.131	0.465	0.500	0.515	0.550

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1SMU and LS1SMA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.37
Voltage withstand - volts, root mean square, minimum	
Conductor to shield.....	5000
Shield to shield.....	500
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Characteristic impedance - ohms at 1 MHz.....	50 ± 2
Group B:	
Attenuation - at 400 MHz, decibels per 100 feet, maximum.....	18
Cold bending, cable - at 0 ± 2°C, 6-inch maximum diameter mandrel.....	No damage
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) type LS1SMU only - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor - conformance to material, construction and coverage, type LS1SMA only	No failure
Shield - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
(B) Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.

UNIT ORDERING LENGTH: 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-1005-40)

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:10Z
Check the source to verify that this is the current version before use.